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Robust bionic distributed multimodal flexible sensor for extreme-condition sensing and intelligent operation

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Traditional multimodal flexible sensors struggle with system integration, limited node scalability, and overall robustness, posing multiple critical challenges. Inspired by the tiger-shark scalp, we present BDMFS, a robust bionic distributed multimodal flexible sensor, integrating an S-shaped optical network mimicking subcutaneous mechanoreceptors with a self-powered triboelectric interface emulating ampullae-based proximity sensing. A microstructured elastic dielectric layer serves as both optical substrate and triboelectric layer, providing exceptional flexibility, mechanical robustness, and environmental adaptability under diverse conditions. BDMFS enables spatiotemporally synchronized perception of proximity (~ 100 mm) and tactile (~ 5 ms) stimuli, detecting gentle touches of 0.25 g while withstanding 6.26 MPa pressures. Coupled with machine-learning, it achieves 95.26% object-proximity recognition accuracy, demonstrated in real-time virtual music teaching, adaptive grasping under low light, and wrist-mounted underwater teleoperation, highlighting its potential for intelligent control and advanced human-robot interaction in extreme environments.

Embarking on the quest for universal human-machine symbiosis^{1,2}, enabling machines to perceive and interpret complex environments with the spatiotemporal fidelity of biological organisms remains a central challenge for next-generation intelligent systems. Flexible sensors, prized for their mechanical compliance and diverse transduction capabilities^{3–9}, have become a cornerstone of wearable electronics^{10,11}, intelligent robotics^{12,13}, and advanced human-machine interfaces^{14–16}. Yet conventional single-point or single-modality devices are increasingly inadequate for real-world, unstructured human-machine-environment interactions because they typically operate reliably only under laboratory conditions and struggle to deliver high-fidelity, time-synchronized decoding of tactile, proximity, and other multimodal cues when subjected to intense mechanical stress and environmental disturbances.

To overcome these limitations, researchers have attempted to integrate multiple sensing units into a single device^{17–20}. Multimodal decoupled sensing is typically achieved through material differences that enable selective responses, structural designs that reduce crosstalk, and signal-processing or algorithm-based methods to separate overlapping signals. For example, this can be achieved by layering liquid-metal tactile elements with triboelectric²¹ or capacitive proximity sensors or integrating flexible piezoresistive devices with ultrasonic units²². However, such “physical stacking” approaches often

sacrifice integration density and robustness²³, leading to signal crosstalk, electromagnetic susceptibility²⁴, and vulnerability to humidity and noise²⁵. Even with three-dimensional microstructures²⁶ or deep-learning-based post-processing^{27,28}, the resulting systems remain complex, less wearable, and algorithmically opaque, falling short of truly synchronous and resilient multimodal perception. Fundamentally, these architectures rely on discrete components in which structure and sensing are separated, yielding signals that are temporally coupled but spatially incoherent.

Nature offers a blueprint to resolve this impasse^{29–31}. As shown in Fig. 1a, during evolutionary adaptation, tiger sharks have developed a highly robust, spatially distributed multimodal sensory network: their head skin combines dynamic compliance with exceptional mechanical strength, while subcutaneous tactile nerve endings are seamlessly integrated with the ampullae of Lorenzini, specialized electroreceptive organs capable of detecting weak near-field electrical signals^{32–34}. This intimate fusion of structure and function embodies the principle of “sensing as structure, structure as sensing”, providing a paradigm for artificial systems that demand simultaneous robustness and synchronous perception.

Inspired by this biological archetype, we present a bionic distributed multimodal flexible sensor (BDMFS). This device co-embeds a distributed optical-fiber network, emulating tactile nerves, and self-powered

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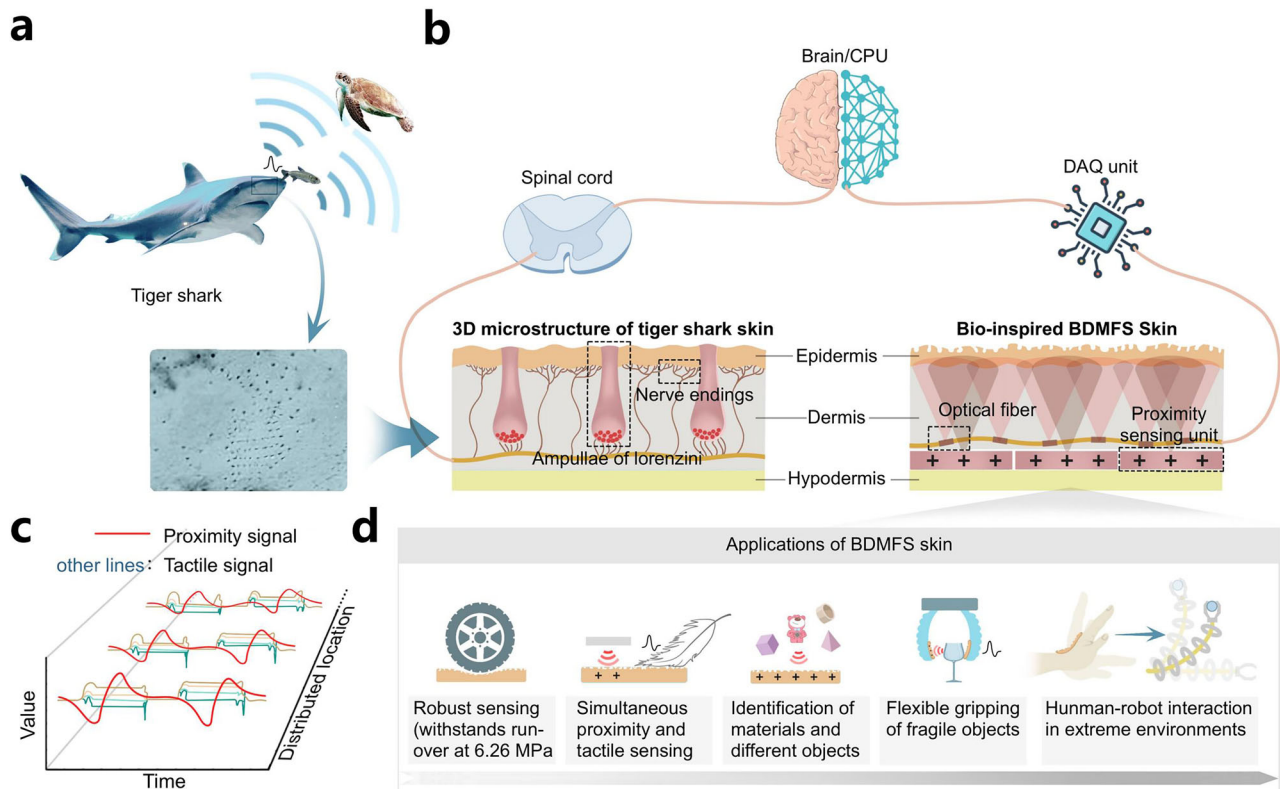


Fig. 1 | Bionic distributed multimodal flexible skin (BDMFS) inspired by tiger-shark sensory mechanisms and its multifunctional applications. **a** Tiger shark hunting its prey through distributed tactile and proximity sensing. (Shark element photo by Zander R on Unsplash; sea turtle element photo by Wexor Tmg on

Unsplash.) **b** Comparison of the sensory mechanisms between tiger-shark head skin and the bionic flexible skin BDMFS. **c** Schematic diagram of the distributed signals output by the bionic flexible skin BDMFS. **d** Multi-scenario, multifunctional applications enabled by the bionic flexible skin BDMFS.

triboelectric proximity units, analogous to Lorenzini ampullae, within a single elastic matrix (Fig. 1b). A microstructured dielectric surface serves concurrently as the optical fiber host and the triboelectric active layer, forming a function-structure integrated platform capable of inherent spatial decoupling and temporal synchronization of optical and electrical signals under severe mechanical shock and harsh environmental conditions. As a result, unlike existing triboelectric-optical hybrid systems^{35–37}, BDMFS achieves high flexibility, environmental adaptability, and mechanical endurance, sustaining loads up to 6 MPa while preserving performance, resolving micro-pressures as low as 0.25 g, enabling 100 mm-scale proximity mapping, and delivering distributed tactile responses within 5 ms-performance levels unmatched by existing flexible multimodal devices (Supplementary Table 6, Supplementary Information).

Leveraging these capabilities, BDMFS performs spatiotemporally synchronized decoding of proximity and tactile signals (Fig. 1c), referring to the simultaneous response of multiple sensing modalities at different spatial locations within a single, non-array device to external stimuli, rather than a sequential, multi-stage sensing process. When combined with data-driven algorithms, it reaches recognition accuracies of 95.26% for object approach detection and 98.94% for tactile localization when combined with data-driven algorithms. As shown in Fig. 1d, demonstrations in virtual music instruction (Supplementary Video S7), flexible grasping of transparent objects under in low-visibility conditions (Supplementary Video S8), and remote control of an underwater cable-driven robotic arm (Supplementary Video S9) underscore its broad potential in human-machine interaction, soft robotics, and operations in extreme environments.

This work provides a solution capable of achieving spatiotemporally synchronized cross-modal sensing in conventional multimodal flexible sensors and resolves their persistent stability challenges under extreme mechanical stress and complex environmental disturbances. Through a bio-inspired function-structure integrated design, the BDMFS achieves the

BDMFS achieves distributed multimodal sensing, which maintains dynamic proximity detection and millisecond-level tactile response even under ultra-high-pressure conditions, and has been successfully validated across multiple scenarios and functionalities, providing a practical and scalable pathway toward highly reliable flexible sensing, adaptive soft robotics, and resilient human-machine interfaces in complex environments.

Results

Working principle and decoupling mechanism

Tiger sharks rely on two complementary distributed sensory systems to achieve seamless tracking from long-range detection to close-range targeting during predation (Fig. 1a). Their entire skin is covered with dermal denticles enriched with mechanoreceptive nerve endings, forming a distributed tactile network capable of sensing subtle hydrodynamic disturbances and minute pressure variations, thereby enabling precise localization of prey motion and position. Concurrently, the head region is densely populated with ampullae of Lorenzini (pore-like electroreceptors) that detect the weak bioelectric fields generated by muscular contractions and cardiac activity of hidden prey, allowing near-field localization even in dark or turbid waters^{38,39}. Working in concert, these dual modalities provide the tiger shark with highly resolved spatial information and tactile feedback in the final moments of approach and capture, ensuring exceptional hunting efficiency. Inspired by this sensory paradigm, we designed a bionic sensing system (BDMFS) that integrates a distributed optical network mimicking mechanoneural tactile perception with a self-powered triboelectric proximity unit emulating the ampullae of Lorenzini, all embedded within a single conformal elastomeric layer patterned with microstructures (Fig. 1b).

In detail, the BDMFS sensor consists of four principal functional layers (Fig. 2a): a flexible dielectric layer, an optical-fiber sensing layer, a flexible electrode layer, and a substrate encapsulation layer, which together emulate the structure of biological epidermis, dermis, and subcutaneous tissue. To

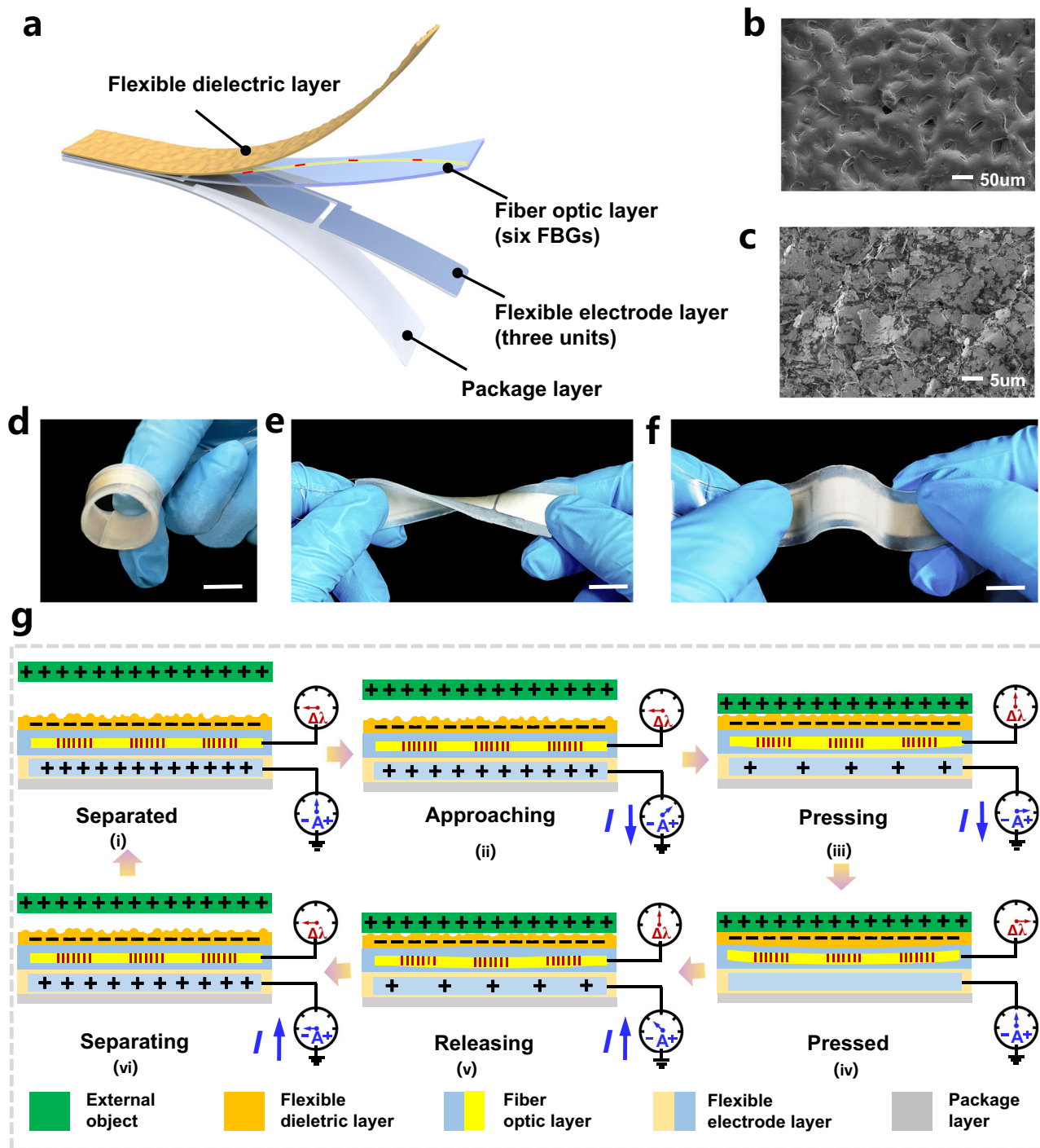


Fig. 2 | Structure and working principle of the bionic distributed multimodal flexible sensor (BDMFS). **a** The BDMFS structural composition. **b** Surface SEM image of the flexible dielectric layer. **c** Surface SEM image of the flexible electrode layer. **d** BDMFS in a curled state. **e** BDMFS in a torsional state. **f** BDMFS in a wrinkled state. (Scale bars in Fig. 2d–f represent 10 mm.) **g** Sensing mechanism: (i) Different electron affinities between materials result in balanced surface charges at the dielectric layer (green) object (orange) interface. (ii) Approaching without

contact causes electrons to flow from ground to the flexible electrode. The optical fiber (yellow) remains unstressed (zero wavelength shift). (iii) Initial contact enhances electron transfer and induces fiber deformation, resulting in a wavelength shift. (iv) Full contact leads to charge neutralization and maximum shift. (v) Pressure release causes electron backflow and gradual signal recovery. (vi) Full release restores the fiber signal to baseline.

enhance the triboelectric charge generation of the dielectric layer, a rough surfaced texture was introduced using replica molding on sandpaper templates, eliminating the need for costly microfabrication methods such as pyramid structuring. The flexible dielectric layer was fabricated from Eco-flex-0030, a skin-like silicone elastomer known for its softness and stretchability. The morphology of the flexible dielectric layer is shown in the

scanning electron microscopy (SEM) image in Fig. 2b. To mimic the distributed nerve endings of skin, a dense array of fiber Bragg gratings (FBGs) was inscribed on 120 μm microfibers using 100 nJ, 200 ns femtosecond laser pulses, achieving a balanced combination of structural robustness and sensitivity. The central wavelengths of the FBGs are listed in Supplementary Table 1 (Supplementary Information). These were subsequently embedded

within the elastic silicone to form the optical fiber sensing layer for tactile detection, organically integrated with the dielectric layer (see Supplementary Fig. 1a in the Supplementary Information for the optic fiber sensing mechanism). On the surface of a pre-stretched 90 μm thick thermoplastic polyurethane (TPU) film (maximum elongation up to 500%), a 1×3 array of block scale-like silver-paste flexible electrodes was fabricated via repeated screen-printing processes to capture proximity-induced charges with high sensitivity, emulating the ampullae of Lorenzini for distributed proximity sensing. The printed electrode layer was then cured at 120 $^{\circ}\text{C}$ for 15 min in a temperature-controlled oven, and its SEM image is displayed in Fig. 2c. The detailed fabrication workflow and the BDMFS prototype are provided in Supplementary Fig. 2 (Supplementary Information). As shown in Fig. 2d, e, BDMFS exhibits skin-like mechanical compliance, capable of withstanding 360 $^{\circ}$ bending, 180 $^{\circ}\text{C}$ twisting, and nearly 90 $^{\circ}\text{C}$ creasing without performance degradation, demonstrating excellent conformability and compatibility for integration with complex surfaces.

As illustrated in Fig. 2g, the working mechanism of BDMFS can be divided into multiple stages: (i) In the initial stage, due to differences in electron affinity between materials, full contact between the external object (orange) and the flexible dielectric layer (green) leads to the generation of equally distributed positive and negative charges at the interface. (ii) As the object approaches the flexible dielectric layer without making physical contact, free electrons are driven from the ground toward the flexible electrode, while the optical fiber (yellow) remains unstrained, resulting in zero wavelength shift. (iii) Upon contact between the external object (green) and BDMFS, electron transfer is enhanced and the embedded optical fiber undergoes mechanical deformation, producing a measurable wavelength shift. (iv) Once full contact is established, charge neutralization occurs, halting further electron flow, and the wavelength shift of the fiber reaches its maximum value. (v) During the release phase, as the external pressure is relieved, free electrons flow back from the flexible electrode (light blue) to the ground, leading to a gradual reduction in wavelength shift as the sensing channel returns toward its initial state. (vi) When the external object is fully detached from BDMFS, the backflow of electrons increases and the optical fiber's wavelength returns to baseline.

As shown in Fig. 3a, b, the BDMFS sensing system is capable of simultaneously detecting wavelength shifts $\Delta\lambda_{\text{total}}$ induced by both tactile and temperature, as well as electrostatic induction voltage $V(t)$ associated with proximity signal (PS). Given that the PS is inherently insensitive to temperature variations while exhibiting instantaneous responsiveness to tactile (as discussed in Section 2.2), we define the contact state of BDMFS based on the condition that the absolute value of the tactile signal (TS) exceeds a preset threshold and a transient $V(t)$ is detected. Conversely, the system is considered to be in a non-contact state when this condition is not met. In the non-contact state, BDMFS primarily functions as an ambient temperature and proximity sensor. In contrast, during the contact state, where both thermal and mechanical stimuli coexist, we adopt a commonly used dual-FBG temperature compensation strategy⁴⁰. In this strategy, a mechanically isolated FBG sensor adjacent to the BDMFS serves as a temperature reference, sharing its field and demodulation system for effective environmental compensation. (see Supplementary Fig. 1b in the Supplementary Information for BDMFS-based sensing system). TS is extracted by subtracting the wavelength shift of the ambient temperature signal (ATS) $\Delta\lambda_T$ from the overall sensing information (OSI) $\Delta\lambda_{\text{total}}$ of the embedded FBGs in BDMFS, thereby enabling effective decoupling of temperature and mechanical stress contributions. Simultaneously, the triboelectric signal allows real-time detection of surface disturbance signals (DS), such as slippage and vibration, during contact events. Similarly, the decoupling process of sensing signals during multitask operation of the BDMFS sensing system is illustrated in Supplementary Fig. 3a and Supplementary Note 2 (Supplementary Information).

To validate the decoupling capability of BDMFS under complex thermo-mechanical coupling conditions, a coordinated temperature-force loading platform was constructed (Supplementary Fig. 3b, Supplementary Information). In this setup, BDMFS was placed on a heating stage, with

pressure applied via a transparent polymethyl methacrylate (PMMA) block and a 60 g metal chain. The chain was tensioned using a string routed through a fixed pulley and connected to a lead screw-driven linear slider, allowing dynamic control of the applied load. To ensure localized thermal stability, a perforated cover was used to enclose the sensing region (setup details and signal evolution are shown in Supplementary Note 1 (Supplementary Information)). During the experiment, the temperature was gradually increased from 24.5 $^{\circ}\text{C}$ to 50 $^{\circ}\text{C}$, with a control accuracy of ± 0.5 $^{\circ}\text{C}$. As heating and load reduction progressed via gradual slider movement, we observed a decreasing TS response despite the continuous temperature rise, indicating the thermal independence of TS. Once the temperature stabilized at 50 $^{\circ}\text{C}$ and the pulling motion ceased, TS plateaued, showing only a slight negative wavelength shift, which is attributed to residual compressive stress under non-ideal loading. Upon releasing the string and thereby increasing the applied pressure, the TS signal rose again, with the final wavelength shift deviating by approximately 10 pm from the initial loading state. After complete unloading, both TS and PS signals returned to their baseline values, while the FBG signal continued to reflect temperature changes, demonstrating that BDMFS can function as a non-contact temperature sensor under steady thermal conditions. At the beginning, when the PMMA block was moved rapidly toward and away from the BDMFS surface, only the PS signal was activated, with no corresponding TS response. This result further confirms the distinct and independent behavior of TS and PS in contact and non-contact scenarios. Throughout the entire experimental process, PS was only triggered during approach or touch events and remained unaffected by temperature. In contrast, TS was effectively decoupled from thermal interference through the dual-FBG compensation approach. These results collectively demonstrate the robust signal decoupling performance of the BDMFS system in multimodal sensing under complex thermal environments.

Optimization and sensing characteristics

Surface morphology of the flexible dielectric layer influences charge accumulation density, thereby affecting proximity sensing performance. To optimize the surface structure of the dielectric layer in BDMFS, we employed replica molding using sandpapers with varying grit sizes (80, 150, 320, and 1200 grit) to fabricate flexible dielectric surfaces with different roughness levels (Supplementary Fig. 4, Supplementary Information). Based on the controlled variable method, we fabricated proximity sensing units with different surface textures (Supplementary Fig. 5, Supplementary Information). As shown in Supplementary Fig. 6 and Supplementary Video S1, we conducted a linear sliding rail-based proximity test and recorded the temporal proximity signals (Fig. 4a). Figure 4b reveals that, when a transparent acrylic plate is positioned 80 mm above the BDMFS surface, the proximity sensing units with smoother surface textures generate higher output voltages. However, when the grit number exceeds 320, further improvements in output voltage become negligible (Supplementary Fig. 7, Supplementary Information). Therefore, sandpaper with 320 grit was selected as the optimal template for surface patterning. Unlike conventional approaches that embed optical fibers into elastomers in linear^{41–43}, curved^{43,44}, or random sinusoidal forms⁴⁵, our design adopts a minimum bending radius (≥ 10 mm) and optical signal attenuation (-30 dBm) as boundary conditions. Considering the overall dimensions of the flexible device, we theoretically derived and established an S-shaped embedded optical fiber layout with a clear analytical function expression ($y = 8\sin(0.1(x + 14))$). (See Supplementary Information Figs. S8–S9 for details of the design process). This design not only ensures layout consistency and manufacturability but also enhances the mechanical flexibility of the fiber optic layer. Six equally spaced FBGs were inscribed along the S-shaped fiber to emulate discrete tactile receptors found in tiger shark skin. As shown in Fig. 4c, all FBGs exhibited optical signal attenuation levels greater than -30 dBm, indicating good demodulation capability.

We also found that the thickness of the dielectric layer is a critical parameter. If it is too thin, the optical fiber becomes vulnerable to mechanical damage, if too thick, the sensor loses wearability. To ensure

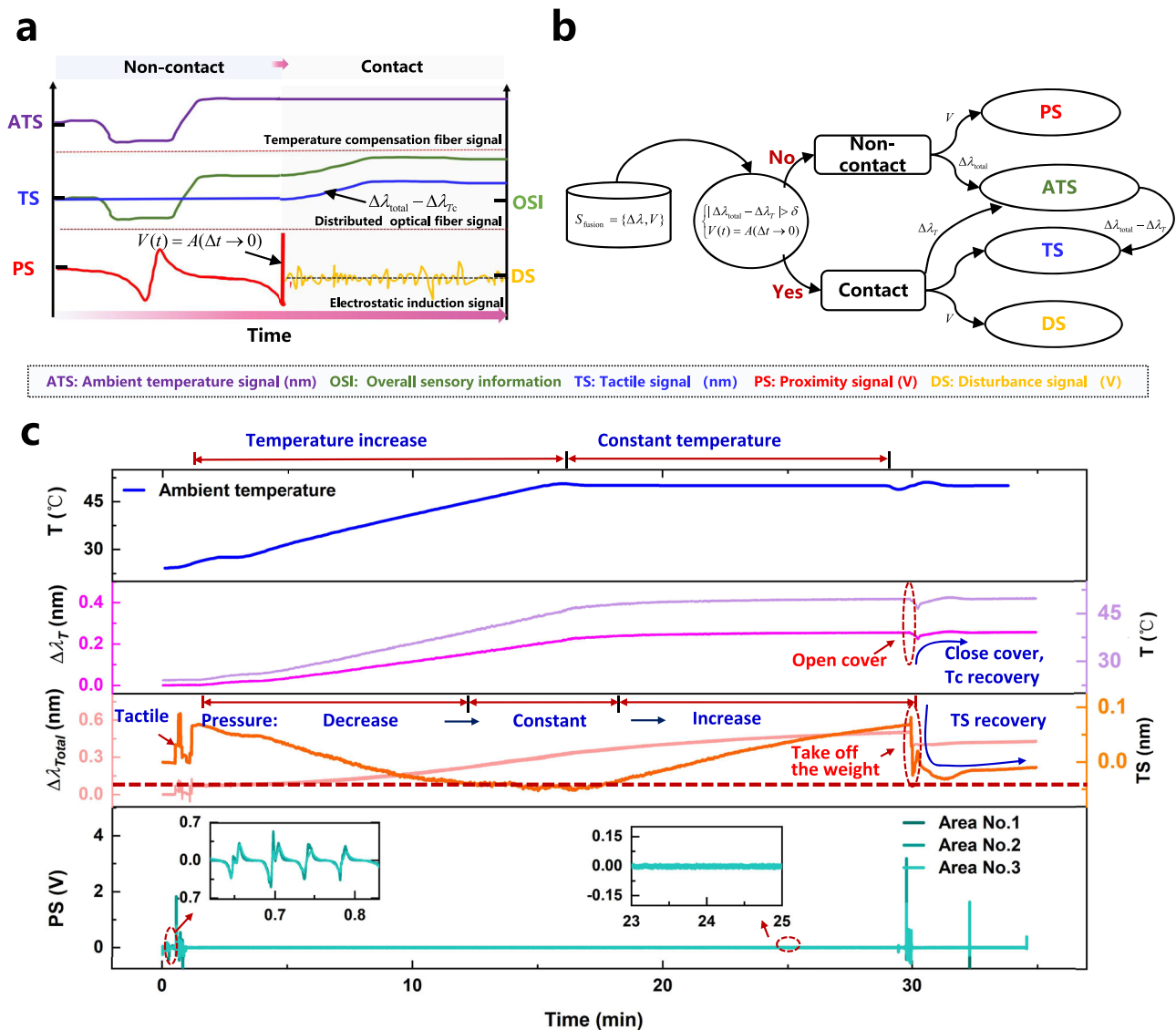


Fig. 3 | Signal Decoupling Mechanism of the bionic distributed multimodal flexible sensor (BDMFS) for multimodal sensing under contact and non-contact Conditions. a Schematic of BDMFS Multimodal Perception in Contact and Non-

Contact States. **b** Mechanism of Multimodal Signal Separation and Decoupling. **c** Experimental Validation of Signal Decoupling under Wide-Range Temperature Variations (24.5–50 °C).

structural safety, we performed finite element analysis (FEA) in Abaqus on BDMFS with different encapsulation thicknesses (see Figs. S10–S13, Supplementary Information), applying a 10 kPa surface load (~ 9 N) representing typical operating pressure. The maximum stress on the optical fiber remained well below its yield strength in the gigapascal range, confirming structural integrity under normal use. Furthermore, as illustrated in Fig. 4d, increasing the thickness of BDMFS led to a gradual rise in fiber stress, thereby enhancing force sensitivity. Meanwhile, the maximum displacement of the fiber remained relatively unchanged, suggesting that structural thickness had minimal influence on displacement. Considering both sensing performance and wearability, a 3 mm-thick silicone encapsulation layer was ultimately selected. In addition, the FEA results revealed that stress concentration in the fiber occurred at the bending inflection point (corresponding to FBG_2), where the axial stress component E_{11} dominated. To further validate the rationality of the BDMFS design, a comparative analysis between FEA simulations and experimental results was conducted, as shown in Fig. 4e. The results indicated that the wavelength shift increased with applied pressure, and the simulated tactile signals closely matched the experimental data (see Figs. S14–S16 for detailed procedures).

Figure 4f presents the proximity sensing behavior when a PMMA cube approaches region 1 of BDMFS at a constant speed of 20 mm s^{-1} from a distance of 100 mm. As the object moves within 5 mm, the proximity signal gradually decreased (negative polarity) and rapidly reversed (positive polarity) upon retraction. Additionally, distinct signal profiles are observed when withdrawing different materials from BDMFS, attributed to variations in electron affinity (Fig. 4g), providing valuable cues for material recognition. To evaluate pressure sensitivity, forces of 0–5 N were applied at the BDMFS center using a 10 mm circular indenter (Supplementary Fig. 28a, Supplementary Information). The indenter covered FBG_3 and FBG_4, yielding linear responses, while the other FBGs outside the contact area showed weak, nonlinear responses with a maximum shift of 0.071 nm at 5 N (Supplementary Fig. 28b, Supplementary Information), much lower than FBG_3 (0.264 nm) and FBG_4 (0.215 nm) (Fig. 4h). Accordingly, only FBG_3 and FBG_4 are shown with linear fits. As shown in Fig. 4i, the device displayed a linear response to applied pressure, with FBG_3 and FBG_4 exhibiting sensitivities of 52 pm N^{-1} and 44 pm N^{-1} , respectively. The tactile sensors also demonstrated excellent repeatability and minimal hysteresis ($< 10 \text{ pm}$) (Supplementary Fig. 17, Supplementary Information). The thermal response of the device was also investigated. As illustrated in Fig. 4j,

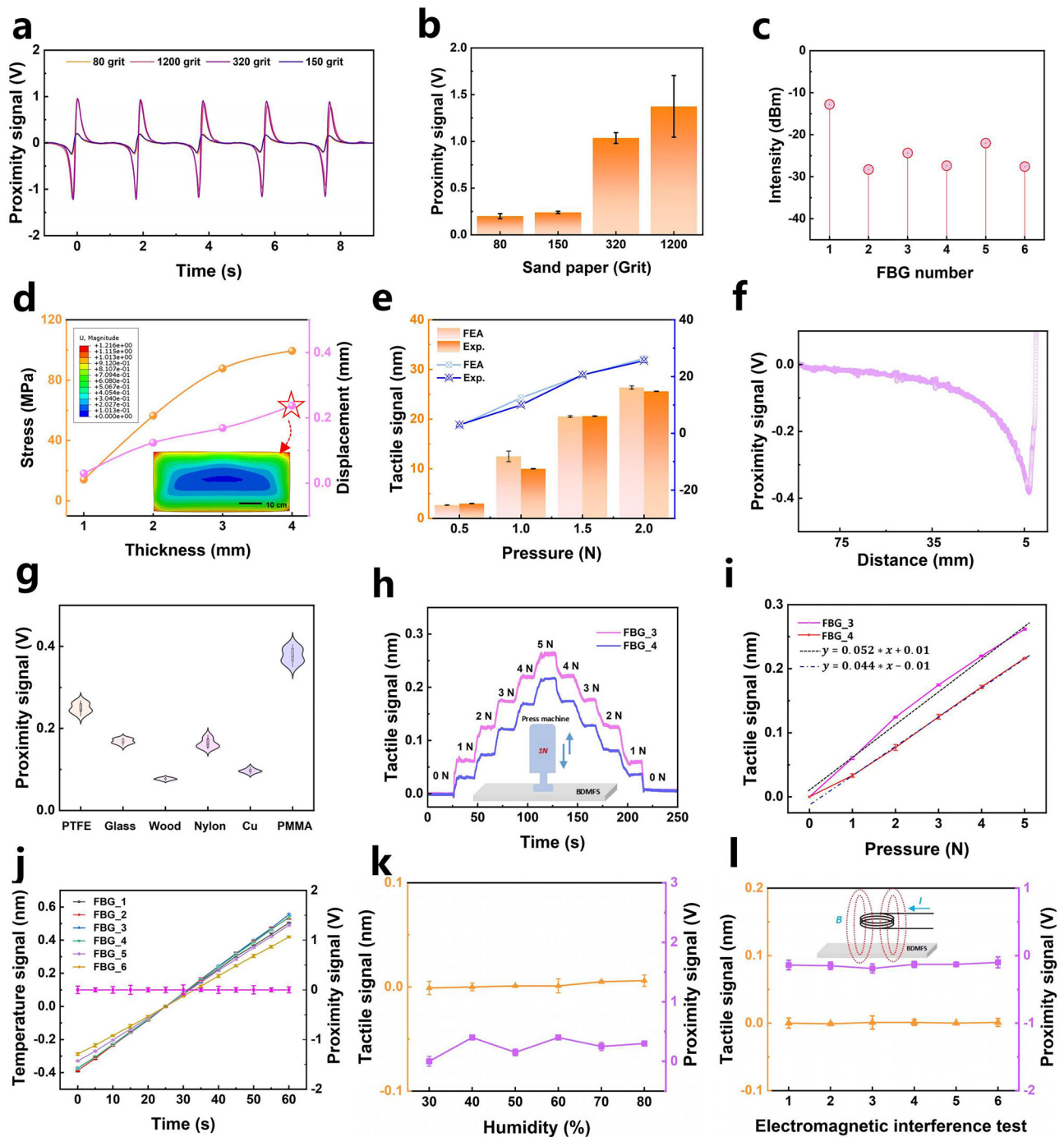


Fig. 4 | Performance characterization of the bionic distributed multimodal flexible sensor (BDMFS). **a** The proximity signal generated by proximity sensing units with different surface morphologies during repeated approach and separation processes with a polymethyl methacrylate (PMMA) block. **b** Maximum voltage output of the tactile units with different surface morphologies measured at a distance of 100 mm from the PMMA. **c** Optical power intensity output when the optical fiber is embedded in an S-shaped configuration within the BDMFS. **d** Maximum stress and displacement changes of the optical fiber under 0.1 MPa pressure for BDMFS with different thicknesses. (U: Total displacement magnitude). **e** FEA and experimental comparison of wavelength shifts at the point of maximum fiber strain (E_{11}). **f** Proximity signal output of BDMFS when a transparent acrylic material approaches at a constant speed of 20 mm s^{-1} **g** Violin plot statistics of the maximum proximity

signal (absolute values) generated by different materials. (PTFE, polytetrafluoroethylene). **h** Cyclic pressure sensing performance of BDMFS using a 10 mm circular indenter applied at the center of the sensor surface with forces ranging from 0 to 5 N. **i** Relationship between tactile signal output (wavelength shift) and applied pressure. **j** Tactile and proximity responses of BDMFS under varying temperature conditions. **k** Tactile (e.g., FBG_1) and proximity (e.g., region 1) signal outputs under varying humidity conditions. **l** Tactile and proximity signal responses of BDMFS to electromagnetic interference induced by a Tesla coil pulse generator (operating at 150 MHz, 120 W) positioned 3 cm from the sensor surface. Experimental data with error bars were obtained from three independent experimental repeats.

while the proximity signal remained stable with increasing temperature, the embedded FBG-based tactile sensors exhibited a linear shift in Bragg wavelength ($R^2 = 0.99$) (Figs. S18–S19, Supplementary Information), confirming the ability of BDMFS to mimic human thermoreception and perform non-contact temperature detection. Supplementary Table 3 (Supplementary Information) summarizes the sensitivities of the FBGs. Under isothermal conditions, the wavelength noise of FBG_6 was measured to be 0.488 pm. Given its temperature sensitivity coefficient of $12 \text{ pm } ^\circ\text{C}^{-1}$, the non-contact temperature sensing resolution of BDMFS is estimated to be approximately $0.04 \text{ }^\circ\text{C}$ (Supplementary Fig. 20, Supplementary Information), which is higher than that of existing multimodal non-contact temperature sensors^{24,46}. To evaluate the influence of humidity on the performance of BDMFS, we conducted environmental tests by placing the sensor in a temperature-controlled chamber maintained at $25 \text{ }^\circ\text{C}$ under varying relative humidity levels ranging from 30 to 80%. The results revealed that fluctuations in humidity had negligible impact on both the tactile and proximity sensing signals of BDMFS (Fig. 4k). Furthermore, electromagnetic interference (EMI) immunity was assessed using a Tesla coil-based electromagnetic pulse generator (operating at 150 MHz and 120 W), positioned 3 cm above the BDMFS surface. As shown in Fig. 4l, BDMFS demonstrated robust electromagnetic immunity, with no signal degradation observed.

Interestingly, even when BDMFS is cut in half, its distributed structure enables the remaining portion to maintain its sensing capabilities (see Supplementary Figs. 21–S22a, Supplementary Information). To verify its mechanical strength, the BDMFS cut in half was subjected to four repeated runs of compression under a 2.37-ton vehicle (Huawei AITO M5, Fig. 5a), with each pass generating a compressive stress of approximately 6.26 MPa (Supplementary Video S2). The results show that BDMFS remained highly sensitive to tactile signals induced by the compression, exhibiting a maximum wavelength shift of 0.693 nm (Fig. 5b). Microscopic inspection of the micro/nanostructured fiber at the most heavily loaded point (FBG3) revealed no mechanical damage (Supplementary Fig. 22b, Supplementary Information). This performance far exceeds the strength limits of conventional flexible sensors, demonstrating its exceptional structural toughness and reliability. In addition, we mounted the sensor on a Fin-Ray soft robotic finger affixed to a UR3 robotic arm^{47,48}. The device underwent repetitive pressing cycles ($\approx 1 \text{ N}$) for approximately 2 hours, totaling over 3500 cycles. Since the fingertip tactile receptors correspond to FBG_5 and FBG_6, which experience the highest concentrated forces, signals from these two channels were selected for acquisition. As shown in Fig. 5c, we observed slight baseline drift in tactile signals during loading/unloading cycles, mainly due to micro-plastic deformation in the dielectric layer and hysteresis from delayed interfacial stress release. However, the tactile response remained generally stable, underscoring the long-term mechanical reliability of BDMFS. Thanks to the integrated structural and functional design and optimization strategies, these comprehensive results highlight the BDMFS's tiger-shark-inspired distributed intelligent sensing capabilities and its resilient, skin-like robustness under extreme environments and mechanical stress, providing compelling evidence for its potential applications in future human-machine interaction scenarios.

Smart perception and human-robot flexible interaction system

Due to the integration of an array of proximity sensing units and micro optical fiber-based tactile receptors within BDMFS, it also possesses multimodal position sensing capabilities. As shown in Fig. 5d, when a feather is gently and uniformly swept across the surface of BDMFS (Supplementary Video S3), BDMFS demonstrates strong spatiotemporal position resolution (Fig. 5e). Based on the time-sequenced signal, the sweeping speed can be accurately calculated as $L/(T_i - T_{i-1}) \approx 29.44 \text{ mm s}^{-1}$ (L is the array spacing, $L = 26.5 \text{ mm}$). We placed a transparent object (PMMA material) above BDMFS and sequentially moved it closer to BDMFS at different positions (Supplementary Video S4). After repeated verification, we found that BDMFS could accurately detect the proximity position of the transparent object (signal strength $> 0.1 \text{ V}$), without generating cross-talk signals

(Fig. 5f). The jitter in the proximity signal may be caused by the unstable motion of the human hand. The proximity sensing ability of BDMFS arises from the difference in the electron affinity between BDMFS and the approaching object. This principle enables BDMFS to identify materials with different properties, such as in automated sorting tasks. Traditional single-point multimodal sensors often fail to detect proximity once contact occurs. Leveraging the distributed architecture of BDMFS, simultaneous sensing of tactile and proximity stimuli is achieved. As shown in Fig. 5g, when a PMMA object approaches region 1 while a finger presses another sensing region, the BDMFS simultaneously detects both stimuli. The response amplitudes of each modality vary with the applied stimuli, enabling synchronized multimodal sensing in complex interaction scenarios. As shown in Fig. 5h, we selected six 30 mm cubic cubes made of different materials: polytetrafluoroethylene (PTFE), glass, wood, nylon, copper, and PMMA. These cubes were fixed onto a moving platform (Supplementary Fig. 6, Supplementary Information), and a linear rail was used to move them towards BDMFS at a constant speed (20 mm s^{-1}) for real-time data collection (Fig. 5i). Each cube was tested 30 times, generating a total of 180 data points. For classification tasks with small data volumes and low feature dimensions, we applied a grid search method to optimize hyperparameters and developed a Support Vector Machine (SVM) classification model (training/validation: 80%/20%) (see Methods section). As demonstrated in Fig. 4j, BDMFS achieved an average accuracy of 95.26% in material identification, with lower accuracy (around 85%) for glass and nylon cubes. This is due to the high similarity in the proximity signals between glass and nylon. In addition to accuracy, class-wise performance metrics provide complementary evidence of model robustness. As summarized in Supplementary Table 4 (Supplementary Information), the precision, recall, and F1-score across different material classes remain consistent and physically meaningful.

BDMFS demonstrates exceptionally fast collaborative sensing capabilities. For instance, when a transparent object, which is visually undetectable, falls from BDMFS (Fig. 6a and Supplementary Video S5), BDMFS first detects the object's proximity within the proximity sensing range (Fig. 6b). Upon contact, BDMFS responds with a remarkably fast reaction time of approximately 1 ms to sense contact and disturbances, with a peak response time around 5 ms (Fig. 6c). We applied a 1 g standard weight to press on the third region of BDMFS, repeating the pressure three times, as shown in Fig. 6d. BDMFS is highly sensitive to micro-stress touches, with the maximum sensitivity reaching up to 20 pm g^{-1} (FBG_3) (Fig. 6e). The sensitivity differences among FBGs are mainly due to the S-shaped layout and local structural variations. We also used objects of varying weights to stimulate BDMFS, including a pad (0.25 g), rubber band (0.34 g), screw (1.16 g), feather (0.28 g), and weight (2 g), to observe its sensitive tactile sensing ability (Supplementary Video S6). Meanwhile, the three proximity sensing regions exhibit exceptional sensitivity to pressure. However, unlike proximity sensing signals, the tactile response in these regions is characterized by pulse signals (Supplementary Fig. 23, Supplementary Information). Due to the distributed tactile receptors (FBGs) in BDMFS, we can precisely identify multiple touch points through the nonlinear tactile sensing characteristics enabled by the S-shaped fiber optic layout. To demonstrate the potential of BDMFS in interactive applications such as flexible musical teaching system, we divided the device into eight finger-sized pressing regions, corresponding to eight musical notes (1-do, 2-re, 3-mi, 4-fa, 5-sol, 6-la, 7-si, 8-do (high)), as shown in Fig. 6f. Random force pressing data were collected from each region (Fig. 6h), with 50 repetitions per region, yielding a total of 400 datasets. As shown in Fig. 6g, a musical note recognition framework based on a deep learning neural network (See Methods section) was developed, achieving a recognition accuracy of 98.94% (Fig. 6i). In addition, Supplementary Table 5 (Supplementary Information) shows consistently high macro-averaged precision (0.961), recall (0.951), and F1-score (0.946), with low inter-fold variance (< 0.035), indicating stable generalization performance. This framework enables real-time identification of the pressed regions on BDMFS, supporting musical note performance (Supplementary Video S7).

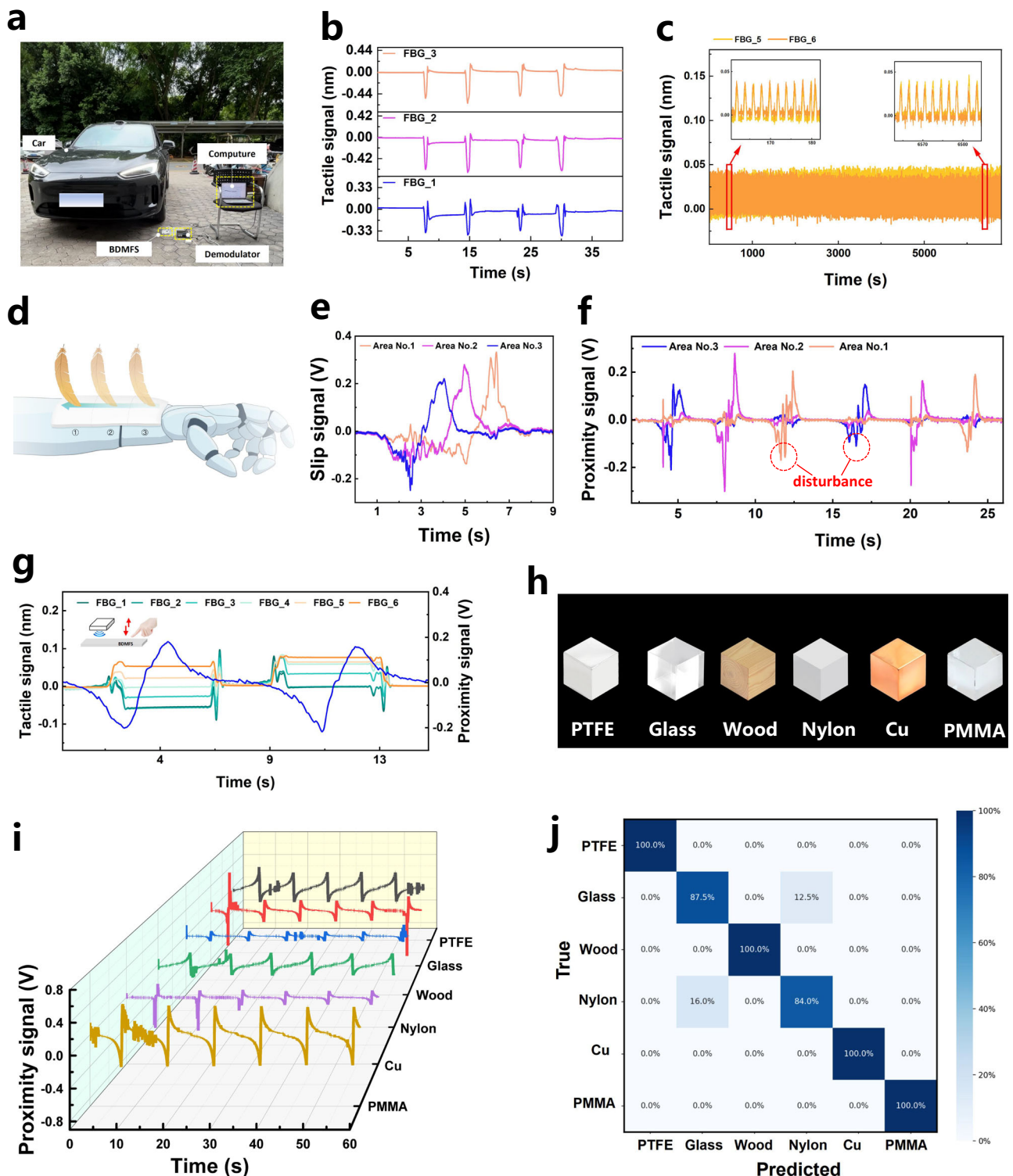


Fig. 5 | Performance and material identification of the bionic distributed multimodal flexible sensor (BDMFS). **a** BDMFS tested under the rolling of a 2.3-ton car. **b** Data collected over four back-and-forth rolling cycles. **c** Cyclic fatigue testing of the BDMFS (e.g., FBG_5 and FBG_6). **d** Gentle stroking of the BDMFS surface with a feather. **e** Tactile sensing signals under the condition of (b). **f** Proximity signal outputs of the BDMFS in response to a transparent glass block approaching different

sensing regions. **g** Simultaneous tactile and proximity sensing. **h** Cubes fabricated from various materials (edge length: 30 mm). **i** Proximity signal responses of the BDMFS when different cubes approach and withdraw at a speed of 20 mm s^{-1} . (PTFE, polytetrafluoroethylene; PMMA, polymethyl methacrylate). **j** Confusion matrix for material classification using a support vector machine (SVM) algorithm, achieving an average recognition accuracy of 95.26%.

To verify BDMFS's capabilities in the synergistic interaction between tactile and proximity sensing, we integrated it into a pneumatic soft robotic gripper, as shown in Fig. 7a. Utilizing the coordinated operation between the UR3 robotic arm and the soft gripper (with the working area of BDMFS

designated as region 1), we successfully performed an approach-grasp-move-release task with a transparent glass cup (Supplementary Video S8). As depicted in Fig. 7b, c, the glass cup movement task was divided into four stages: (1) At stage (i) (Before 7.9s), the pneumatic soft hand gradually

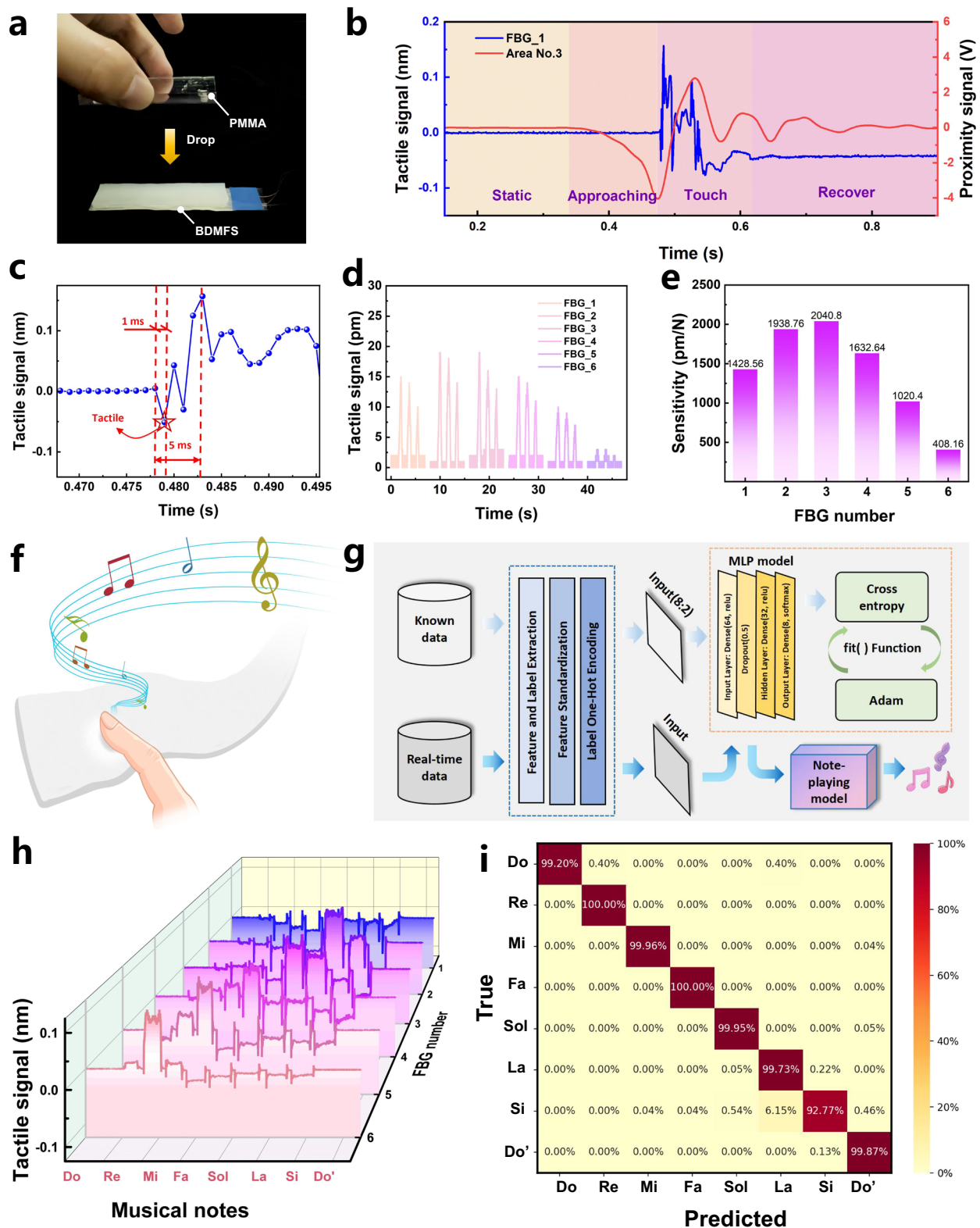


Fig. 6 | Tactile perception and position recognition of the bionic distributed multimodal flexible sensor (BDMFS). **a** A transparent acrylic block is dropped from a height of 10 cm above the BDMFS surface. **b** Tactile and proximity response curves of the BDMFS under the condition of (a). **c** Tactile response time of the BDMFS. **d** A 1g standard weight is pressed three times onto Region 3 of the BDMFS. **e** Maximum sensitivity analysis of tactile receptors within the BDMFS under the

condition of (d). **f** Musical note playing using the BDMFS as a flexible electronic skin. **g** Framework of the deep learning-based algorithm for musical note recognition with BDMFS. **h** Pressing signals collected from different position regions of the BDMFS. **i** Confusion matrix of note recognition performance, with an average recognition accuracy of 98.94%.

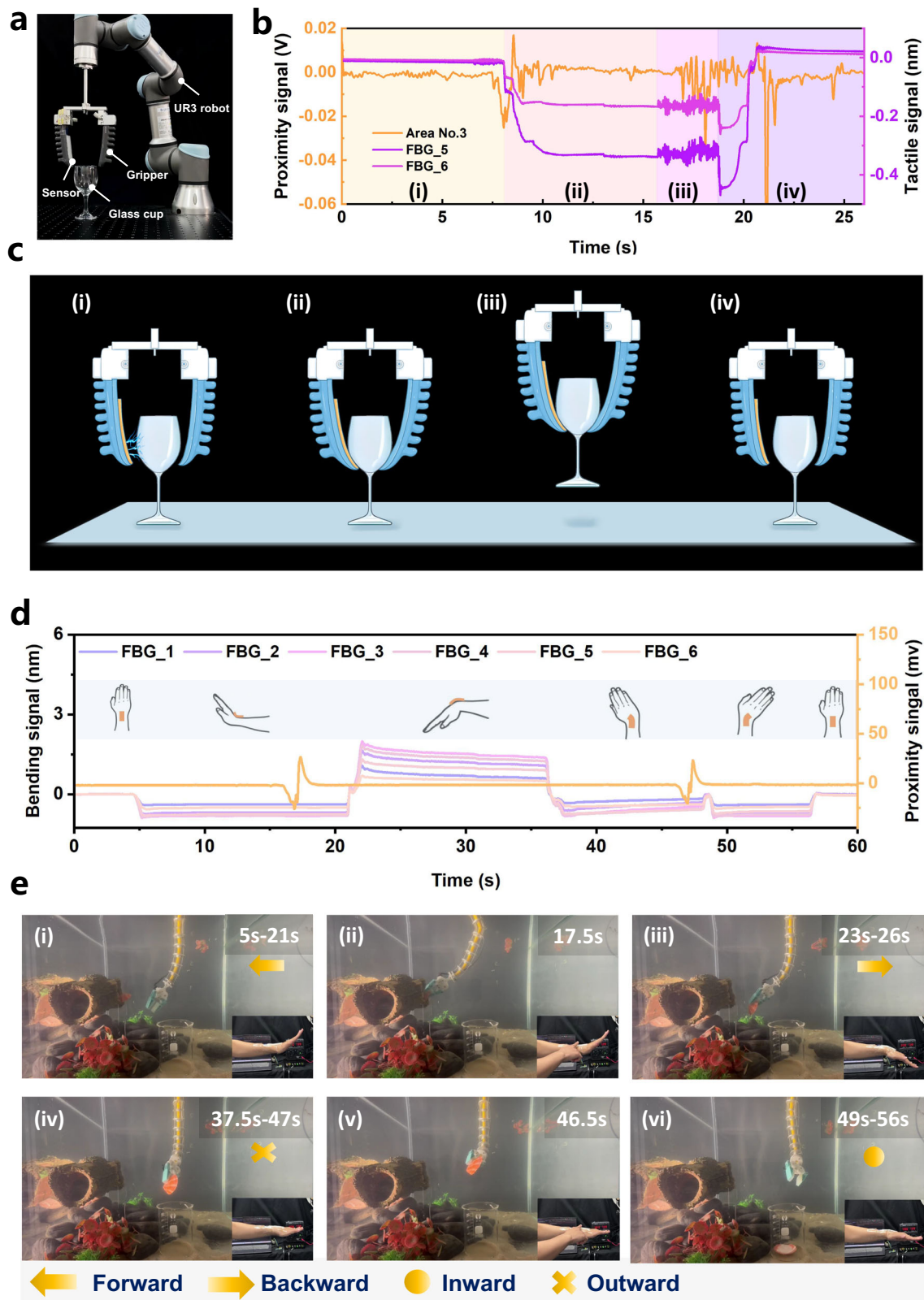


Fig. 7 | Flexible manipulation tasks enabled by the bionic distributed multimodal flexible sensor (BDMFS). **a** A soft robotic gripper equipped with the BDMFS sensor grasping a transparent glass cup. **b** Output signals from the BDMFS during the grasping operation of the soft robotic gripper. **c** Key operational states of the soft robotic gripper during the manipulation process. **d** Real-time signal data from BDMFS during gesture-based teleoperation of an underwater cable-driven flexible robotic arm. **e** Key operational states of the underwater soft arm: (i) Slow wrist flexion from rest (5 s–21 s), causing forward motion of the soft arm. Upon reaching

the target location, proximity sensing via the fingers triggers the soft gripper to close and grasp a shell (at 17.5 s); (ii) Wrist extension (23 s–36 s) drives the soft arm to move backward, with the shell continuously held by the gripper; (iii) Wrist ulnar deviation (37.5 s–47 s) causes the soft arm to move outward; (iv) Upon reaching the designated position, proximity sensing again triggers the gripper to release the shell into an underwater beaker (at 46.5 s); (v) Wrist radial deviation (49 s–56 s) results in inward movement, with the system returning to its initial state after 56 s.

approached the glass cup. During this stage, the fingers closed slowly as driven by air pressure, bringing the hand closer to the glass cup (7.5 s to 7.9 s). (2) At stage (ii) (7.9 s–15.6 s), the soft robotic hand made contact with the glass cup and, with continued air pressure, tightened its grip. At 15.6 s, the air pressure reached its peak value, and at this moment, the tactile sensing from BDMFS reached its maximum (the FBG wavelength shift was at its highest). (3) At stage (iii) (15.6 s–18.7 s), while the soft hand maintained its grip on the glass cup, the UR3 robotic arm began to execute slow horizontal movements. During this phase, the gripping force of the soft hand remained constant (as the driving air pressure remained unchanged), so BDMFS's tactile perception remained stable without proximity sensing signals. During the arm's movement, there was a little jitter due to the angular movement of the robotic arm and the elastic material of the fingers, which affected the downward motion of the glass cup. Both tactile and proximity signals exhibited fluctuations. Despite the lack of visual assistance, we were still able to accurately discern the real presence of proximity sensing through tactile feedback. (4) At stage (iv) (18.7 s–20.6 s), the bottom of the glass cup made contact with the table, generating a large impact force. This moment caused an enhancement in the tactile signal from BDMFS. From 18.7 to 20.6 s, the soft hand gradually released the glass cup, with the grip strength diminishing. During this release, the tactile signal from BDMFS weakened, and after 20.6 s, once the soft hand fully released the cup, both tactile and proximity signals ceased. This experiment demonstrates that BDMFS can enhance the intelligent perception and operational capabilities of the soft robotic gripper, particularly in tasks involving the grasping of transparent objects or in low-visibility environments where visual feedback is unavailable.

To further explore the potential of BDMFS in human-robot interaction within complex environments such as narrow underwater spaces, we have developed a dual-flexible teleoperation system, which integrates the flexible BDMFS sensor with a flexible cable-driven arm. By wearing BDMFS on the wrist and leveraging perceptual information from various hand gestures, we remotely controlled a cable-driven underwater flexible arm (Supplementary Fig. 24, Supplementary Information). Based on our previous work⁴⁹, this cable-driven underwater flexible arm, using deep reinforcement learning strategies, can perform detection and interaction in underwater complex environments. It can autonomously generate underwater movement trajectories to achieve predefined objectives, and use its end effector, a soft gripper, to capture delicate objects. Leveraging this system, we implemented remote operation of the underwater flexible arm using wrist movements. Specifically, the wrist's static, extension, flexion, ulnar deviation, and radial deviation were mapped to corresponding movements of the end effector of the cable-driven flexible arm: static, forward, backward, outward, and inward, respectively. Proximity signals controlled the opening and closing of the end effector's soft gripper. BDMFS's receptors exhibit nonlinear characteristics in response to different hand gestures. For nonlinear target classification, deep learning offers robust advantages. Consequently, we constructed a deep neural network model (See Methods section) for real-time gesture recognition (algorithm framework shown in Supplementary Fig. 25, Supplementary Information). We collected data for each gesture 50 times, totaling 250 datasets. After five-fold cross-validation, in which the dataset was randomly divided into five subsets, with 80% of the data used for training and 20% for validation in each fold, and the results averaged across folds with standard deviations computed to evaluate model generalization, this model achieved an accuracy of up to 100% (Supplementary Figs. 26–S27, Supplementary Information). Beyond overall accuracy, macro-averaged precision, recall, and F1-score are used to evaluate model performance. (Supplementary Table 6, Supplementary Information) Five-fold cross-validation yields values of 0.972, 0.969, and 0.969, respectively, with low cross-fold variability (< 0.02) indicating stable generalization across different data splits. Figure 7d, e demonstrate the remote operation of the flexible arm based on BDMFS and gesture commands (Supplementary Video S9). The task includes real-time shell grasping and targeted placement in confined spaces: (i) The wrist slowly moves from a static position to flexion (5 s–21 s), causing the flexible arm to move forward. Upon reaching

the designated position, the proximity signal from the fingers is used to control the soft gripper to close and pick up the shell (17.5 s). (ii) The wrist extends (23 s–36 s), causing the flexible arm to move backward, while the soft gripper holds onto the shell. (iii) The wrist performs ulnar deviation (37.5 s–47 s), causing the flexible arm to move leftward. (iv) After reaching the designated position, the proximity signal from the fingers is used to release the shell into the underwater beaker (46.5 s). (v) The wrist performs radial deviation (49 s–56 s), causing the flexible arm to move rightward, returning to the initial state at 56 s. Through this task, we confirmed that BDMFS, with its multimodal sensory capabilities, enables human-robot flexible interaction in various extreme scenarios (not limited to underwater environments). Compared to rigid remote operation devices, BDMFS offers higher flexibility and control in operation.

Discussion

To address the limitations of current multimodal flexible sensors, namely their single-point sensing and insufficient robustness, which hinder performance in complex human-machine-environment interactions, We developed the BDMFS, a robust BDMFS with excellent perceptual capabilities. The embedded S-shaped optical fiber and the micro-structured dielectric elastomer layer together establish an integrated sensing-load-bearing physical framework. The S-shaped fiber, through geometric bending and reserved buffer zones, substantially enhances its stretchability within the elastomer, maintaining linear and repeatable wavelength responses under large deformations. Meanwhile, the microstructured dielectric layer not only boosts triboelectric charge generation and collection efficiency but also serves as the direct mechanical coupling medium for the optical fiber, enabling spatially coordinated tactile and proximity signal distribution. This bionic “structure-as-function” strategy enhances durability while mitigating fatigue fracture at multilayer interfaces, constituting a key origin of the sensor's mechanical robustness. Notably, although existing multimodal sensor arrays can achieve spatiotemporally synchronized perception, they typically rely on independently operating, spatially arrayed nodes and system-level timing control. In contrast, the proposed BDMFS adopts a bio-inspired unified structural design and mechanical transmission pathway, which tightly couples all sensing modalities, leveraging the distributed sensing and wavelength-division multiplexing features of FBGs to enable relatively high-density tactile sensing with reduced wiring complexity. With the integration of heterogeneous proximity sensing units, the system can be simultaneously activated under diverse physical interactions, thereby offering improved integration, compatibility, and environmental robustness compared to conventional array-based schemes.

As shown in Fig. 8 and Supplementary Table 7 (Supplementary Information), comparison with representative multimodal (tactile/proximity) sensors^{1,21–25,46,50–52} indicates that the BDMFS exhibits superior overall performance, primarily attributed to its bioinspired function-structure integrated design. We found that BDMFS also exhibits exceptional precision and long-term stability in multimodal signal detection. Statistical analyses across multiple testing batches reveal that, under identical conditions, the tactile sensing variation remains within 10 pm with a standard deviation below 5 pm; proximity signal noise is maintained below 0.06 V; and temperature sensing errors remain within 10 pm. Vertical force-loading tests (Supplementary Fig. 28, Supplementary Information) confirm a linear correlation ($R^2 = 0.99$) between the applied pressure and wavelength shifts, with measurement errors confined to within 100 mN (Supplementary Table 8, Supplementary Information). Under controlled conditions, BDMFS was subjected to three-week repetitive testing of tactile, proximity, and thermal sensing (Supplementary Figs. 29–S31, Supplementary Information), showing consistent performance with tactile and thermal signal deviations within 10 pm, and a maximum proximity signal deviation of only 4.93%, confirming excellent long-term reliability. In proximity sensing experiments involving complex objects (Supplementary Figs. 32–S34, Supplementary Information), it was found that the magnitude of BDMFS's proximity sensing is related to the object's shape, surface texture, and surface material. This is attributed to the fact that proximity sensing is governed by

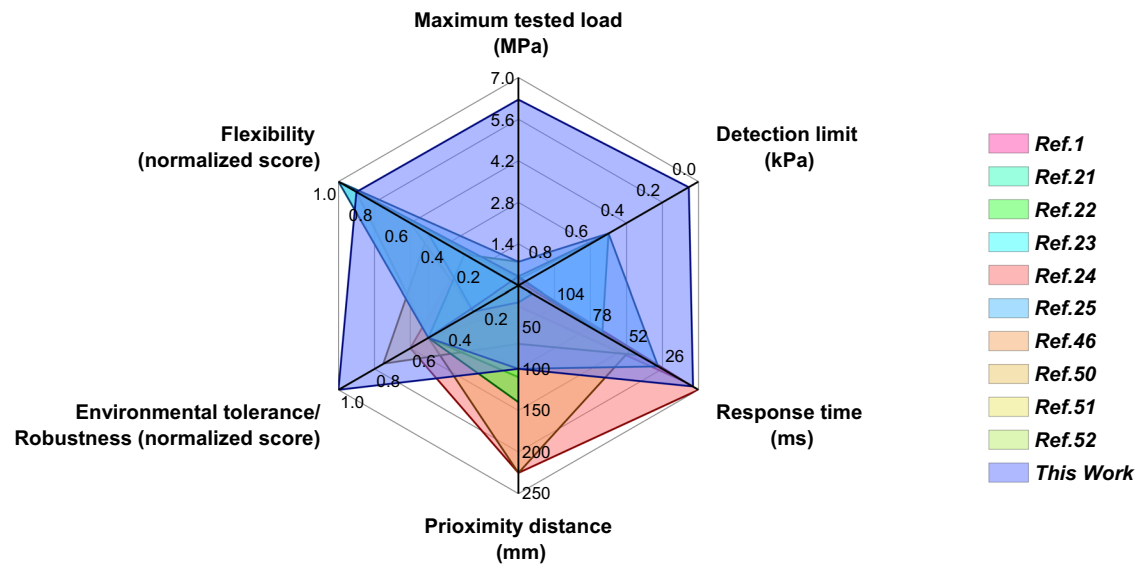


Fig. 8 | Comparison between this work and other reported studies. Detailed normalization rules are described in Supplementary Note 3 (Supplementary Information).

differences in electron affinity between the object and the sensor surface. Therefore, surface contamination or coatings may alter the polarity or magnitude of the triboelectric charge, leading to changes in the proximity response. It should be noted that triboelectric proximity sensing may be affected by environmental humidity and variations in target surface conditions. These effects can be mitigated through structural encapsulation and bioinspired superhydrophobic dielectric layers, together with threshold-based or algorithmic compensation strategies, thereby enhancing system stability and robustness.

Furthermore, BDMFS maintains good linearity and adaptability under mechanical deformation, such as bending (Supplementary Fig. 35, Supplementary Information) and stretching (Supplementary Fig. 36, Supplementary Information). The device can accurately detect deformation-induced wavelength shifts and rapidly recover to its initial state within 2 seconds after unloading, highlighting excellent mechanical resilience. Importantly, proximity signals remain unaffected by such deformations. This stability is largely attributed to the S-shaped configuration of the embedded optical fiber, which increases the fiber's stretchability within the elastomeric layer by nearly tenfold. It is worth noting that under complex loading conditions involving simultaneous stretching, bending, and contact, signal decoupling becomes increasingly difficult or even infeasible due to superimposed multimodal responses. Nevertheless, this complexity endows BDMFS with a rich perceptual interface and functional redundancy, making it suitable for applications in sophisticated human-machine interaction scenarios. For instance, in fruit-sorting and gesture-recognition tasks involving soft robotics, BDMFS sustains a 100% gesture-recognition accuracy even when up to three tactile sensing nodes fail (Supplementary Fig. 37 and Supplementary Table 9, Supplementary Information). This highlights how its distributed architecture not only strengthens reliability but also creates fertile ground for future integration of machine learning and advanced signal-decomposition algorithms. It is worth noting that classification on small-scale datasets may risk overfitting. To mitigate this, we use cross-validation, model complexity control, early stopping, and data standardization. Moreover, the multimodal signals for different materials, notes, and gestures are physically well separated, which reduces classification ambiguity under limited-sample conditions.

These findings demonstrate that BDMFS leverages a bionic distributed architecture and structurally integrated multimodal sensing design to address the limitations of traditional point-based flexible sensors in environmental adaptability and functionality. This work represents an advance in multimodal sensing, shifting from single-point to distributed mapping and from simple signal superposition to functional decoupling. It lays a solid

foundation for next-generation human-machine collaborative perception systems that are highly robust, multidimensional, and scalable, and potential applications in intelligent robotics, soft manipulation, and operations in extreme environments.

Conclusion

Inspired by the sensory system of tiger-shark skin, we developed a robust BDMFS. The design integrates a distributed optical sensing network, mimicking mechanoneural pathways, with a self-powered triboelectric unit inspired by the Lorenz ampullae, all embedded within a single, conformal elastomeric layer. Through a function-structure co-design strategy, the elastomeric dielectric layer, patterned with surface microstructures, simultaneously serves as both the optical fiber matrix and the triboelectric interface, imparting exceptional flexibility, environmental adaptability, and mechanical robustness.

Benefiting from this structural paradigm, BDMFS delivers excellent comprehensive performance. It achieves distributed multimodal sensing, offering an array-based proximity detection range of 100 mm and a distributed tactile response time of 5 ms, together with high sensitivity (52 pm g^{-1}) and a reliable dynamic tactile response under high-stress conditions. The device readily distinguishes minute pressures down to a 0.25 g standard weight while withstanding extreme loads of 6.26 MPa under repeated vehicle-wheel rolling without performance degradation, maintaining long-term stability across 3500 cycles. Leveraging the synergistic effect of multimodal perception, BDMFS accurately identifies contact states and effectively decouples tactile, proximity, and non-contact temperature signals. Owing to its distributed tactile architecture, the system retains 100% gesture-recognition accuracy even when three sensing units are deliberately disabled.

Furthermore, when combined with data-driven algorithms, BDMFS attains recognition accuracies of 95.26% for object-proximity detection and 98.94% for tactile localization. Demonstrations including a virtual music-teaching interface, flexible grasping of transparent objects under vision-limited conditions, and remote manipulation of an underwater cable-driven robotic arm underscore its broad application potential in human-machine interaction, soft robotics, and extreme-environment operations. Notably, BDMFS also supports multi-scenario functional reuse: in a non-contact mode it monitors ambient temperature with a fine resolution of 0.04°C , while under uniaxial stretching it can sense bending states, offering a feasible approach for robotic joint-flexion monitoring. These results collectively confirm that BDMFS surpasses existing multimodal sensors in both performance and application breadth. In summary, this study provides an

effective strategy to spatiotemporal synchronization of cross-modal information in conventional multimodal flexible sensors and enhances their stability under extreme mechanical stress and complex environmental disturbances. Multiscenario, multifunctional experiments further demonstrate their strong potential for intelligent interaction in extreme conditions.

Despite its excellent distributed multimodal perception performance, simultaneous exposure to stretching, compression, bending, and thermal variations can generate complex signal patterns that may require advanced decoupling algorithms. Future research will therefore focus on integrating additional sensing modalities and vision-based technologies to further advance intelligent multimodal interaction systems.

Methods

Relevant theory

Assuming the optical fiber embedded within the elastic layer of the BDMFS follows a functional expression $f(x)$, the S-shaped fiber layout should satisfy:

$$\begin{cases} k(x) = \frac{|f''(x)|}{(1+(f'(x))^2)^{3/2}} \leq \frac{1}{10} \\ P_{\text{FBG}_i} > -30 \text{ dBm}, i = 1, 2, \dots, 6 \end{cases} \quad (1)$$

where $k(x)$ denotes the curvature of the curve, and P_{FBG_i} represents the reflected optical power of each FBG.

The BDMFS system determines a contact state based on Equation (2): contact is established when the tactile sensing signal exceeds a predefined threshold δ , while the proximity sensing channel outputs a narrow pulse signal A. Otherwise, the system is considered to be in a non-contact state.

$$\begin{cases} |\Delta\lambda_{\text{total}} - \Delta\lambda_T| > \delta \\ V(t) = A(\Delta t \rightarrow 0) \end{cases} \quad (2)$$

Under contact conditions, the wavelength drift of the FBG $\Delta\lambda_T$ induced by temperature variations can be calculated:

$$\Delta\lambda_T = \frac{\Delta\lambda_{\text{comp}}}{k_{\text{comp}}} \cdot K \quad (3)$$

where $\Delta\lambda_{\text{comp}}$ denotes the wavelength shift of the compensation FBG, k_{comp} is the temperature sensitivity coefficient of the compensation FBG, and K refers to the temperature sensitivity coefficient of the non-compensated FBG.

Mold manufacturing

The molds for the flexible dielectric layer and fiber optic sensing layer of BDMFS were fabricated using 3D printing. The molds were printed with ABS material using a 3D printer (X1-Carbon, Shenzhen Tuozhu Technology Co., Ltd.).

Material preparation

The flexible dielectric layer and fiber optic sensing layer of BDMFS were prepared using Ecoflex-0030 (Smooth-On, USA) material (detailed process shown in Supplementary Fig. 2, Supplementary Information). The electrode layer was made from TPU material (tensile strength 35 MPa, elongation at break greater than 450%) (Xinrui Shop, Taobao, China), with silver paste (Shenzhen Yilai Technology Co., Ltd., China) used as the conductive layer. The customized formula for the silver paste is provided in Supplementary Information Supplementary Table 2. The proximity sensing array electrodes were fabricated using a custom screen printing machine. Curing of the materials was performed using a temperature-controlled drying oven (Dongguan Zhicheng Co., Ltd., China). The adhesion between Ecoflex and TPU is achieved using Wacker (Germany) E43 adhesive.

BDMFS performance characterization

Pressure calibration was performed using a push-pull force gauge (HANDPI, China) mounted on a manual screw rack (HLA). Sensitivity

testing utilized a standard 1g calibration weight (C-type, Ouniu Co., Ltd.). Proximity sensing was tested using a precision ball screw linear guide (SFU1610, Zhejiang Hengxin Co., Ltd.) with a stroke of up to 500 mm. Temperature measurements were calibrated using a temperature chamber. The morphology of BDMFS, after gold sputtering, was scanned using a laboratory scanning electron microscope (magnification: 20 – 2000 ×, SU8010).

Proximity sensing signal acquisition

The three-channel proximity signals were connected to a custom-designed signal acquisition card (MircoTENG), which transmitted data to a PC via Wi-Fi. Two data sampling methods were used: one with LabVIEW software, sampling at 1 kHz, and another with a custom Python-based data acquisition system, where the sampling frequency is adjustable.

Tactile and thermal sensing signal acquisition

The optical fiber sensors were connected to a demodulator, which was then linked to the PC via a USB serial connection. Data acquisition was performed through serial communication with the UDP/IP transmission protocol. Similar to the proximity sensing, two sampling methods were employed: one using LabVIEW software at 1 kHz and another using the custom Python-based system with a user-defined sampling frequency.

Classification models

(1) Material classification: A support vector machine (SVM) was used to perform six-class classification based on a scalar feature extracted from the experimental measurements. All features were standardized using z-score normalization prior to training. To address class imbalance under limited data conditions, the synthetic minority over-sampling technique (SMOTE) was applied to the training set. Both linear and radial basis function (RBF) kernels were evaluated, and the regularization parameter C and kernel coefficient γ were optimized using grid search with 5-fold cross validation. (2) Musical note recognition: The input samples comprise six synchronously acquired tactile sensing channels that capture multiphysical responses during tactile interactions. All features are z-score normalized using training-set statistics and consistently applied during validation and inference. Eight-class labels are encoded in a one-hot format. Feature extraction and classification are implemented using a multilayer fully connected neural network with three hidden layers (64, 32, and 16 neurons) and a Softmax output layer. ReLU activations, L2 regularization, and a dropout rate of 0.5 are employed to alleviate overfitting under limited data conditions. The network is trained end-to-end using the Adam optimizer and cross-entropy loss, with early stopping based on validation loss. Model generalization and robustness are evaluated via 5-fold cross validation. (3) Gesture classification: The wrist posture recognition model is a 1D CNN for six-channel tactile inputs. Raw signals are denoised by threshold filtering and normalized using z-score standardization, with the fitted scaler retained for real-time inference. The network consists of two convolutional feature-extraction blocks (kernel size: 3; channels: 32, 64), each followed by ReLU activation and max pooling, and two fully connected layers for classification (FC: 128; Softmax). Through end-to-end training, the model automatically learns high-dimensional discriminative features by integrating multi-channel tactile information. Model training is performed using 5-fold cross validation (training/validation: 80%/20%), with the Adam optimizer and cross-entropy loss (epochs: 100). As shown in Supplementary Fig. 26 (Supplementary Information), training loss is monitored and confusion matrices are computed for each fold, demonstrating stable feature learning and robust generalization on a limited dataset.

Flexible actuator

The pneumatic soft gripper and its control system (Soft Touch Co., Ltd., China) were used to perform the glass cup grasping task. The underwater flexible arm grasping system is a self-developed device.

Data analysis

Unless otherwise specified, all experiments were repeated multiple times to obtain reliable data. Data analysis was conducted using Origin 2022 software.

Language refinements

Some text was refined using a large language model (LLM), and all changes were checked by the authors to ensure the original meaning was preserved.

Ethics declarations

This study involves only non-invasive wearable sensors. According to the institutional policy, formal ethics committee approval is not required for this type of research.

Data availability

The source data underlying the figures are provided as Supplementary Data. Further information about this study is available from the corresponding author.

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Author contributions

Baijin Mao and Yedong Huang conceived the study and developed the methodology, and also wrote the original draft. Yuyaoen Xiang managed the data, performed formal analysis, and conducted the investigation. Wenbo Liu and Xulong Shi were responsible for validation and reviewed and edited the manuscript. Xiang Qian provided supervision. Juntian Qu administered the project. All authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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